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Auxiliary station cooling unit for DC faster charge

Abstract

A charging station for a vehicle, a system and a method of charging the vehicle. The system includes a heat exchanger at the vehicle. The charging station includes a reservoir including a charging station coolant, a dispenser unit configured to circulate the charging station coolant from the reservoir through the vehicle to absorb heat generated at the vehicle, and a cooling unit for cooling the charging station coolant from the reservoir. Heat generated during the charging of the vehicle is transferred into a vehicle coolant circulating within the vehicle. The heat is transferred from the vehicle coolant to a charging station coolant. The charging station coolant is flowed out of the vehicle.

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Background/Summary

INTRODUCTION

- (1) The subject disclosure relates to charging an electrical vehicle and, in particular, to a system and method for dispersing heat generated during the charging of the vehicle.
- (2) Electric vehicles run off of batteries that need to be charged once their energy is depleted. Charging a battery generates an amount of heat directly related to the rate at which the charging occurs. Without heat mitigation, the charging process can lead to thermal runaway. In order to reduce charging times, there is a desire to increase charging rates leading to higher charging temperatures. Accordingly, it is desirable to provide a method of maintaining the temperature of the battery below a runaway temperate limit during a charging process.

SUMMARY

- (3) In one exemplary embodiment, a method of charging a vehicle is disclosed. Heat generated during the charging of the vehicle is transferred into a vehicle coolant circulating within the vehicle. The heat is transferred from the vehicle coolant to a charging station coolant. The charging station coolant is flowed out of the vehicle.
- (4) In addition to one or more of the features described herein, the method further includes flowing the charging station coolant in a coolant circulation loop that includes a cooling unit outside of the vehicle. The method further includes storing the charging station coolant at an underground

reservoir for ambient cooling. The method further includes coupling a hose to the vehicle, the hose including a connector having a first port and a second port, flowing the charging station coolant into the vehicle via the first port, and flowing the charging station coolant out of the vehicle through the second port. The method further includes switching a configuration of a valve of an internal circulation circuit of the vehicle when the hose is connected to the vehicle. Switching the configuration includes placing a heat exchanger in the internal circulation circuit to allow the heat to transfer from the vehicle coolant to the charging station coolant at the heat exchanger. The method further includes cooling the charging station coolant via one of circulating the charging station coolant through a radiator exterior to the vehicle and transferring the heat from the charging station coolant to a secondary coolant.

(5) In another exemplary embodiment, a charging station for a vehicle is disclosed. The charging station includes a reservoir including a charging station coolant, a dispenser unit configured to circulate the charging station coolant from the reservoir through the vehicle to absorb heat generated at the vehicle, and a cooling unit for cooling the charging station coolant from the reservoir.

(6) In addition to one or more of the features described herein, the cooling unit is located outside of the vehicle. The reservoir is underground. The charging station further includes a hose for coupling the dispenser unit to the vehicle, the hose including a connector with a first port for flowing the charging station coolant into the vehicle and a second port for flowing the charging station coolant out of the vehicle. The charging station further includes an electronic control unit configured to send a command to a valve of an internal circulation circuit of the vehicle when the hose is connected to the vehicle to switch a configuration of the valve. Switching the configuration of the valve includes placing a heat exchanger in the internal circulation circuit to allow the heat from the vehicle coolant to be transferred to the charging station coolant at the heat exchanger. The cooling unit further includes at least one of a radiator for cooling the charging station coolant from the reservoir and a condenser for cooling a secondary coolant, the secondary coolant absorbing heat from the charging station coolant.

(7) In yet another exemplary embodiment, a system for charging a battery of a vehicle is disclosed. The system includes a heat exchanger at the vehicle, a reservoir exterior to the vehicle, the reservoir including a charging station coolant, a dispenser unit configured to circulate the charging station coolant from the reservoir through the vehicle to absorb heat generated at the vehicle, and a cooling unit for cooling the charging station coolant from the reservoir.

(8) In addition to one or more of the features described herein, the reservoir is underground. The system further includes a hose for coupling the dispenser unit to the vehicle, the hose including a connector with a first port for flowing the charging station coolant into the vehicle and a second port for flowing the charging station coolant out of the vehicle. The system further includes an electronic control unit configured to send a command to a valve of an internal circulation circuit of the vehicle when the hose is connected to the vehicle to switch a configuration of the valve. Switching the configuration of the valve includes placing the heat exchanger in the internal circulation circuit to allow the heat from the vehicle coolant to be transferred to the charging station coolant at the heat exchanger. The cooling unit further includes at least one of a radiator for cooling the charging station coolant from the reservoir and a condenser for cooling a secondary coolant, the secondary coolant absorbing heat from the charging station coolant.

(9) The above features and advantages, and other features and advantages of the disclosure are readily apparent from the following detailed description when taken in connection with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Other features, advantages and details appear, by way of example only, in the following detailed description, the detailed description referring to the drawings in which:
- (2) FIG. 1 shows a charging station, in accordance with an exemplary embodiment;
- (3) FIG. 2 shows a charging park including a plurality of charging stations, in an embodiment;
- (4) FIG. 3 shows a close-up view of a connector of a dispenser unit of the charging station and a dock of a vehicle;
- (5) FIG. 4 shows a schematic view of a cooling system of the vehicle, in an embodiment;
- (6) FIG. 5 shows a coolant circulation loop of the charging station, in an embodiment;
- (7) FIG. 6 shows a charging station in an alternate embodiment;
- (8) FIG. 7 shows a close-up view of a bottom section of the vehicle;
- (9) FIG. 8 shows a flowchart of a method for controlling a flow of a charging station coolant through the vehicle; and
- (10) FIG. 9 shows a flowchart illustrating a method of cooling the charging station fluid.

DETAILED DESCRIPTION

- (11) The following description is merely exemplary in nature and is not intended to limit the present disclosure, its application or uses. It should be understood that throughout the drawings, corresponding reference numerals indicate like or corresponding parts and features.
- (12) In accordance with an exemplary embodiment, FIG. 1 shows a charging station **100**. The charging station **100** includes a dispenser unit **102** and a hose **104** extending from the dispenser unit **102**. A reservoir **106** storing a charging station coolant is located at the charging station **100**. In various embodiments, the reservoir **106** is an underground reservoir. The dispenser unit **102** circulates cold charging station coolant from the reservoir **106** via the hose **104** into a vehicle **108** parked at the charging station **100**. The charging station coolant from the charging station **100** is delivered to a liquid-to-liquid heat exchanger **120**. The charging station coolant absorbs heat generated during charging of the vehicle's battery. The hot charging station coolant is then circulated from the vehicle **108** back to the reservoir **106** via the hose **104** and dispenser unit **102**, where it is cooled.
- (13) The hose **104** includes a connector **110** at one end, which can include a non-spill connector. The vehicle **108** has a dock **112** for receiving the connector **110**. The connector **110** is placed in the dock **112** so that the charging station coolant can circulate through the vehicle **108** heat exchanger **120** to cool the battery as it is being charged. Although not shown in FIG. 1, the charging station **100** also includes a charging unit and a charging cable for charging the battery of the vehicle **108**. In various embodiments, the charging cable and the hose **104** can be combined into a single hose assembly.
- (14) FIG. 2 shows a charging park **200** including a plurality of charging stations **100a-100e**, in an embodiment. A driver parks the vehicle **108** at a selected charging station (e.g., charging station **100a**) and connects the charging cable and hose **104** (FIG. 1) to the vehicle for charging. During charging, charging station coolant **202** is circulated from the reservoir **106** to the vehicle **108** and back to the reservoir. Each charging station **100a-100e** is in fluid communication with the reservoir **106**. Thus, the charging station coolant can be shared from the reservoir **106** to any of the charging stations **100a-100e**.
- (15) FIG. 3 shows a close-up view **300** of the connector **110** of the dispenser unit **102** and the dock **112** of the vehicle **108**. The connector **110** includes a first port **302** and a second port **304**. The dock **112** includes an input socket **306** and an output socket **308**. When the connector **110** is connected to the dock **112**, the first port **302** is inserted into the input socket **306** to form a sealed inflow passage and the second port **304** is inserted into the output socket **308** to form a sealed outflow passage. Charging station coolant **202** flows into the vehicle **108** via the first port **302** and the input socket **306** and flows out of the vehicle via the output socket **308** and the second port **304**. In various

embodiments, the dock **112** can be located near a charging dock **310** that receives a charging cable. (16) FIG. 4 shows a schematic view **400** of a cooling system of the vehicle **108**, in an embodiment. The schematic view **400** shows an interior region **402** that lies within or is on the vehicle **108** and an exterior region **404** that lies outside of the vehicle. Partition **406** shows a location at which the connector **110** is plugged into the dock **112**. The location at which charging station coolant **202** enters the vehicle **108** is indicated by the first port **302** and the input socket **306** and the location at which coolant exits the vehicle is indicated by second port **304** and the output socket **308**. The charging station coolant **202** enters the vehicle **108** at a cold temperature (e.g., 20 degrees Celsius). The coolant **202** passes through the heat exchanger **120** and exits the vehicle **108** at a high temperature (e.g., 30 degrees Celsius) upon absorbing energy from a vehicle coolant **420**. The heat exchanger **120** is located inside or at the vehicle **108**.

(17) The interior region **402** includes an internal coolant circuit **410**. The internal coolant circuit **410** includes a pump **412**, a valve **414**, a chiller **416**, and a rechargeable energy storage system (RESS), such as a battery **418**. A vehicle coolant **420** is circulated through the internal coolant circuit **410** via the pump **412**. The valve **414** can be switched between a first configuration and a second configuration. In the first configuration, the internal coolant circuit **410** includes the pump **412**, the valve **414**, the chiller **416**, and the battery **418**. In the first configuration, only the chiller **416** is used to cool the vehicle coolant **420**. In the second configuration, the internal coolant circuit **410** includes the pump **412**, the valve **414**, the chiller **416**, the battery **418** and the heat exchanger **120**. The valve **414** is switched to the second configuration during a charging process for the battery to include use of the heat exchanger **120**.

(18) The valve **414** can be switched from the first configuration to the second configuration when the connector **110** is coupled to the dock **112**. Removing the connector **110** from the dock **112** switches the valve **414** back to the first configuration. In one embodiment, the connector **110** applies a force on the valve **414** when it is placed in the dock **112** to mechanically switch the valve from the first configuration to the second configuration. Removing the connector **110** removes the force, thereby allowing the valve to switch back to the first configuration. In another embodiment, the valve **414** can switch to the second configuration upon receiving an electronic signal that is transmitted by the connector **110**. This signal can be transmitted when the connector **110** is secured to the dock **112** or when the connector **110** is within a selected distance of the dock **112** and/or valve **414**.

(19) In the first configuration of the valve **414**, the vehicle coolant **420** is circulated between the battery **418** and the chiller **416** to cool the battery during general operation of the vehicle. When the valve **414** is in the second configuration, the vehicle coolant **420** passes through heat exchanger **120**. The heat generated at the battery **418** during charging is carried via the vehicle coolant **420** to the heat exchanger **120** where it is transferred to the charge station coolant **202**. The charge station coolant **202** is then circulated out of the vehicle **108**, thereby removing the heat from the battery **418** and maintaining the battery **418** at a temperature low enough to prevent overheating (i.e., below a battery runaway temperature).

(20) FIG. 5 shows a coolant circulation loop **500** of the charging station **100**, in an embodiment. The coolant circulation loop **500** includes an un-insulated underground reservoir **502**, an insulated reservoir **504**, and a cooling unit **506**. The charging station coolant **202** returning from the vehicle **108** is deposited into the un-insulated underground reservoir **502**, where its temperature stabilizes to the temperature of the ground. From the un-insulated underground reservoir **502**, the charging station coolant **202** is circulated to either the cooling unit **506** or the insulated reservoir **504** for additional cooling.

(21) The cooling unit **506** includes a first cooling system **508** and a second cooling system **510**. The first cooling system **508** includes a radiator **512** and fan **514**. The second cooling system **510** including a pump **516**, a compressor **518**, a condenser **520** and a chiller **522**. The fan **514** can also be used in combination with the second cooling system **510**. In one configuration, the

charging station coolant **202** from the un-insulated underground reservoir **502** is circulated through the radiator **512** for cooling. The fan **514** blows ambient air over the radiator **512** in order to draw heat out of the charging station coolant **202**. In another configuration, the charging station coolant **202** from the un-insulated underground reservoir **502** is circulated to the insulated reservoir **504** where the heat therein is transferred from the charging station coolant **202** to a secondary coolant **524**. The secondary coolant **524** is then circulated from the insulated reservoir **504** through the second cooling system of the cooling unit **506**, where it is cooled and then returned to the insulated reservoir **504**. The fan **514** can be operated to facilitate condensation in the condenser **520**.

(22) In various embodiments, the secondary coolant **524** can be a phase change material. The phase change material is kept at cold temperature using the second cooling system **510**. In an embodiment, the phase change material is ice/water and the second cooling system **510** maintains the water at about 0 degrees Celsius. The pump **516** circulates the secondary coolant **524** between the insulated reservoir and the second cooling system **510**.

(23) A coolant valve **526** in the coolant circulation loop **500** switches between a first configuration and a second configuration. In the first configuration, the charging station coolant **202** is directed from the un-insulated underground reservoir **502** to the first cooling system **508** of the cooling unit **506**. At the cooling unit **506**, the charging station coolant **202** passes through a radiator **512** where it is cooled. The charging station coolant **202** is then circulated back to the dispenser unit **102** and vehicle. The charging station coolant **202** passes through a filter **528** at the dispenser unit **102**. A dispensing pump **530** at the dispenser unit **102** circulates the coolant **202** into the vehicle **108**, FIG. 1.

(24) In the second valve configuration, the charging station coolant **202** is directed from the un-insulated underground reservoir **502** to the insulated reservoir **504**. At the insulated reservoir **504**, the coolant **202** exchanges heat with a secondary coolant **524** for cooling. The coolant **202** is then send to the dispensing station, passing through the filter **528** and dispensing pump **530**. Meanwhile, the secondary coolant **524** is pumped from the insulated reservoir **504** through the second cooling system **510**, which maintains the temperature of the secondary coolant within a given range of temperatures. A surge tank **540** can be used to handle spillage within the second cooling system **510**.

(25) An electronic control unit **532** is in communication with various components of the coolant circulation loop **500**, such as the fan **514**, the compressor **518**, the pump **516**, the dispensing pump **530** and the valve **526**. The electronic control unit **532** controls the operation of these components in order to maintain operation of the coolant circulation loop **500**. For example, when the ambient temperature is less than a selected temperature, the valve **526** can be placed in a first configuration to direct the coolant to the radiator **512**. When the ambient temperature is greater than or equal to the selected temperature, the valve **526** can be placed in a second configuration to circulate the coolant to the insulated reservoir **504**. In various embodiments, the selected temperature is a room temperature (e.g., 25 degrees Celsius).

(26) In various embodiments, the electronic control unit **532** is also in communication with the valve **414** of the refrigerant circulation loop **410** of the vehicle **108** and can send a control signal to the valve to changes its configuration.

(27) FIG. 6 shows a charging station **600** in an alternate embodiment. The charging station **600** includes a dispenser unit **102** and a cart **602** which can be rolled underneath the vehicle **108** when the vehicle is parked at the charging station. A dock is located at a bottom section **604** of the vehicle **108** for fluid transfer.

(28) FIG. 7 shows a close-up view **700** of the bottom section **604** of the vehicle **108**. The bottom section **604** includes a door **702** that can be opened and closed to reveal or conceal the dock. The door **702** can be a sliding door in various embodiments.

(29) A first hose **704** extends from the cart **602** to the dispenser unit **102** and a second hose **706** extends from the cart to the dispensing system. At the cart **602**, the first hose **704** terminates in a

first port **302** and the second hose terminates in a second port **304**. The door **702** can be opened when the cart is underneath the vehicle **108** to expose an input socket **306** and an output socket **308**. The cart **602** can be arranged so that the first port **302** can be sealingly coupled to the input socket **306** and the second port **304** can be sealingly coupled to the output socket **308**. Coolant **202** can then be pumped from the reservoir (**106**, FIG. **1**) into the vehicle **108** via the first hose **704**, first port **302** and input socket **306**. Coolant is removed from the vehicle **108** and sent to the reservoir (**106**, FIG. **1**) via the output socket **308**, second port **304** and the second hose **706**.

(30) FIG. **8** shows a flowchart **800** of a method for controlling the flow of the charging station coolant **202** through the vehicle. In box **802**, a test is performed to see if the coolant connector is connected to the vehicle. If the connector is not connected to the vehicle, the method proceeds to box **804**. In box **804**, no action is taken and the method returns to box **802**. If at box **802**, the connector is found to be connected to the vehicle, the method proceeds to box **806**. In box **806**, the temperature of the battery is measured. If the temperature is greater than an ambient temperature threshold (e.g., 25 degrees Celsius), the method proceeds to box **808** and box **810**. In box **808**, the valve **414** is placed into the second configuration to include the heat exchanger **120** in the internal coolant circuit. In box **810**, a count is made of the number of vehicles that need cooling at the charging station. In box **812**, the pump rate of the charging station coolant is adjusted to accommodate the number of vehicles that need cooling. In an embodiment, the number of charging vehicles n is multiplied by a standard pump rate for a single vehicle (e.g., 15 liters per minute) to obtain a combined pump rate (e.g., $n \times 15$ liters per minute).

(31) In box **814**, the fan of the charging station is turned on to initiate cooling of the charging station coolant. In box **816**, the mode of operation for cooling the charging station coolant is selected. If the ambient temperature is less than the selected ambient temperature threshold (e.g., 25 degrees Celsius), then the method proceeds to box **818**. Otherwise, the method proceeds to box **820**.

(32) In box **818**, the coolant valve **526** is switched to the first configuration so that the charging station fluid **202** flows through the radiator. In box **820**, the coolant valve **526** is switched to the second configuration so that the charging station fluid **202** flows through the insulated reservoir **504**.

(33) The charging station coolant **202** is therefore cooled based on the ability of the charging station to cool the charging station coolant **202** to the desired temperature. If there is adequate capacity for rejecting heat directly to the air, then the radiator cooling is selected. Otherwise, the use of the chiller is selected. This method of selection cooling methods optimizes the cooling process.

(34) FIG. **9** shows a flowchart **900** illustrating a method of cooling the charging station fluid. In box **902**, a test is performed to determine the ambient temperature. If the ambient temperature is less than the ambient temperature threshold (e.g., 25 degrees Celsius), the method proceeds to box **904**. At box **904**, no action is taken and the method returns to box **902**. If at box **902**, the ambient temperature is greater than or equal to the ambient temperature threshold, the method proceeds to box **906**. At box **906**, a check is made on the temperature of the insulated reservoir or of the secondary coolant. If the temperature of the secondary coolant is less than a phase change temperature (e.g., 0 degrees Celsius), then the method proceeds to box **904** in which no action is taken. If, at box **906**, the temperature of the secondary coolant is less than a phase change temperature, the method proceeds box **908**. At box **908**, the compressor is turned on. At box **910**, the stationary fan is turned on. At box **912**, the pump **516** is turned on to circulate the secondary coolant.

(35) While the above disclosure has been described with reference to exemplary embodiments, it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted for elements thereof without departing from its scope. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the

disclosure without departing from the essential scope thereof. Therefore, it is intended that the present disclosure not be limited to the particular embodiments disclosed, but will include all embodiments falling within the scope thereof.

Claims

1. A method of charging a vehicle, comprising: transferring a heat generated during the charging of the vehicle into a vehicle coolant circulating within the vehicle; transferring the heat from the vehicle coolant to a charging station coolant; flowing the charging station coolant out of the vehicle and into an underground reservoir associated with a charging station; and switching a coolant valve between a first configuration and a second configuration, wherein, in the first configuration, the coolant valve circulates the charging station coolant from the underground reservoir through a radiator for removal of the heat and, in the second configuration, the coolant valve circulates charging station coolant from the underground reservoir to an insulated reservoir for removal of the heat to a secondary coolant fluid stored in the insulated reservoir.
2. The method of claim 1, further comprising selecting between circulating the charging station coolant through the radiator circulating the charging station coolant to the insulated reservoir based on an ambient temperature.
3. The method of claim 2, wherein the charging station further comprises a fan coupled to the radiator and a condenser, further comprising using the fan to cool the charging station coolant at the radiator and to cool the secondary coolant fluid in the condenser.
4. The method of claim 1, further comprising coupling a hose to the vehicle, the hose including a connector having a first port and a second port; flowing the charging station coolant into the vehicle via the first port; and flowing the charging station coolant out of the vehicle through the second port.
5. The method of claim 4, further comprising switching a configuration of a valve of an internal circulation circuit of the vehicle when the hose is connected to the vehicle.
6. The method of claim 5, wherein switching the configuration includes placing a heat exchanger in the internal circulation circuit to allow the heat to transfer from the vehicle coolant to the charging station coolant at the heat exchanger.
7. The method of claim 1, further comprising cooling the vehicle coolant via a chiller in the vehicle.
8. A charging station for a vehicle, comprising: an underground reservoir including a charging station coolant; a dispenser unit configured to circulate the charging station coolant from the underground reservoir through the vehicle to absorb a heat generated at the vehicle; and a cooling unit for cooling the charging station coolant from the underground reservoir, the cooling unit including: a radiator; an insulated reservoir that stores a secondary coolant fluid; and a coolant valve switchable between a first configuration in which the coolant valve circulates the charging station coolant from the underground reservoir through a radiator removal of the heat and a second configuration in which the coolant valve circulates the charging station coolant from the underground reservoir to an insulated reservoir for removal of the heat to the secondary coolant fluid stored in the insulated reservoir.
9. The charging station of claim 8, wherein the coolant valve is placed in one of the first configuration and the second configuration based on an ambient temperature.
10. The charging station of claim 9, wherein the cooling unit further comprises a fan coupled to the radiator and a condenser, wherein the fan cools the charging station coolant at the radiator and cools the secondary coolant fluid in the condenser.
11. The charging station of claim 8, further comprising a hose for coupling the dispenser unit to the vehicle, the hose including a connector with a first port for flowing the charging station coolant into the vehicle and a second port for flowing the charging station coolant out of the vehicle.

12. The charging station of claim 11, further comprising an electronic control unit configured to send a command to a valve of an internal circulation circuit of the vehicle when the hose is connected to the vehicle to switch a configuration of the valve.
 13. The charging station of claim 12, wherein switching the configuration of the valve includes placing a heat exchanger in the internal circulation circuit to allow the heat from a vehicle coolant to be transferred to the charging station coolant at the heat exchanger.
 14. The charging station of claim 8, wherein the charging station coolant is sent from one of the radiator and the insulated reservoir to the vehicle.
 15. A system for charging a battery of a vehicle, comprising: a heat exchanger at the vehicle; an underground reservoir exterior to the vehicle, the underground reservoir including a charging station coolant; a dispenser unit configured to circulate the charging station coolant from the underground reservoir through the vehicle to absorb a heat generated at the vehicle; and a cooling unit for cooling the charging station coolant from the underground reservoir, the cooling unit including: a radiator; an insulated reservoir that stores a secondary coolant fluid; and a coolant valve switchable between a first configuration in which the coolant valve circulates the charging station coolant from the underground reservoir through a radiator for removal of the heat and a second configuration in which the coolant valve circulates the charging station coolant from the underground reservoir to an insulated reservoir for removal of the heat to the secondary coolant fluid stored in the insulated reservoir.
 16. The system of claim 15, wherein the coolant valve is placed in one of the first configuration and the second configuration based on an ambient temperature.
 17. The system of claim 15, further comprising a hose for coupling the dispenser unit to the vehicle, the hose including a connector with a first port for flowing the charging station coolant into the vehicle and a second port for flowing the charging station coolant out of the vehicle.
 18. The system of claim 17, further comprising an electronic control unit configured to send a command to a valve of an internal circulation circuit of the vehicle when the hose is connected to the vehicle to switch a configuration of the valve.
 19. The system of claim 18, wherein switching the configuration of the valve includes placing the heat exchanger in the internal circulation circuit to allow the heat from a vehicle coolant to be transferred to the charging station coolant at the heat exchanger.
 20. The system of claim 15, wherein the charging station coolant is sent from one of the radiator and the insulated reservoir to the vehicle.
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